

OJAS GRAMOPADHYE

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EDUCATION

New York University, New York
Master of Science, Computer Science

Aug '24 - May '26
GPA : 3.89/4

Specialization : AI & ML, Natural Language Processing

Courses : Artificial Intelligence, Deep Learning, Robot Perception, Big Data, Machine Learning, Design & Analysis of Algorithms, Information Visualization, ML Systems Engineering (Spring 2026)

Indian Institute of Technology Bombay, India
Bachelor of Technology, Computer Science and Engineering

May '19 - Jul '23
GPA : 7.74/10

Courses : AI & ML, Speech Recognition, Image Processing, Database Systems, Discrete Mathematics, Data Structures & Algorithms, Software Systems

KEY PROJECTS

Multi-subject alignment : Neural Speech Decoding
Graduate Research | NYU Video Lab

Sept '25 - Ongoing
New York University

- Developing a multi-subject ECoG to speech system that learns subject-invariant embeddings via InfoNCE based contrastive alignment and multi-subject adapters for robust cross-subject phoneme decoding
- Advancing brain-to-speech translation with a spatial-attention electrode mapper using 3D MNI geometry, Conv/GRU encoders, and pretrained ASR supervision (HuBERT/Wav2Vec 2.0) for intelligible speech generation

Modulus : Agentic AI Assistant
Independent Project

Jun '25 - Ongoing
New York University

- Building an agentic assistant using the Model Context Protocol (MCP) and LangGraph to manage multi-threaded conversations, dynamic tool routing, and stateful LLM execution with real-time streaming
- Implementing vector-based memory using ChromaDB & MongoDB – backed persistence for short and long-term context tracking, with tool integration, a scalable FastAPI backend, and a React frontend

Flowsight : NYC Traffic Collision Analysis and Prediction
Course Project | Big Data

Jan '25 - May '25
New York University

- Developed a real-time big data pipeline using Apache Kafka and MongoDB to ingest, validate, and store live NYC collision and traffic speed data; implemented exponential backoff and retry mechanisms for fault-tolerance
- Engineered a Spark-based ML pipeline for crash-risk scoring using spatially matched traffic and collision data, training a Linear Regression model to identify high-risk locations with real-time dashboard visualization using React and Flask

Autonomous Mobility Tool for Visually Impaired
Team Sixth Sense, NYU

Jan '25 - Jun '25
New York University

- Developed a vision-language Scene Description system for aiding the visually impaired by leveraging space-separated scene parsing, and methods like few-shot prompting, Chain of Thought reasoning, and iterative verification for LLMs

WORK EXPERIENCE

Detection and Mitigation of Hallucination in Generative Models | [arXiv](#) 
Research Assistant

May '23 - May '24
IBM Research | IIT Bombay

- Researched hallucination mitigation in generative language models by creating a Subjective Medical QA dataset and developing a differential diagnosis system inspired by real medical practices; published in EMNLP Findings 2024.
- Achieved up to 90% accuracy by implementing a Forward-Backward Chain of Thought method and fine-tuning LLaMA-2-Chat (7B and 70B) models for reward generation with Hugging Face and Accelerate

Japanese OCR Pipeline
Data Science Intern

May '22 - Jul '22
Daikin

- Developed an OCR system using a MobileNetV3 + DB-based text detector on a dataset of 3000 proprietary company documents in Japanese, and achieved a Levenshtein Distance of 0.91 for character recognition

Locust Occurrence Modeling and Prediction
Data Science Associate

May '21 - Jul '21
Dtime.ai

- Predicted locust outbreaks in East Africa by engineering geospatial features and training a Bi-directional ConvLSTM U-Net for pixel-level classification on timestamped 10km² grids, using SMOTE to handle data skew

SKILLS

Focus	Neural Speech Representations, Agentic AI, Natural Language Processing (Modeling, Reasoning & Alignment)
Frameworks	PyTorch, TensorFlow, HuggingFace, PyTorch Lightning, LangGraph, LangChain, OpenCV, DeepSpeed, Accelerate
Systems	Distributed ML (Spark, Dask), Data Engineering (Kafka, MongoDB, PostgreSQL, ChromaDB), MLflow, SLURM HPC
Development	Python, C++, JavaScript, Docker, Node.js, FastAPI, React.js, d3.js